MOLECULAR DYNAMICS SUMMARY

Effect of pH and Salt on the Aggregation of Silica Nanoparticles in Water

CLAYFF + SPC water, LAMMPS | two 3 nm α-quartz nanoparticles | October 2026

Abstract

Two 3 nm crystalline silica nanoparticles were simulated in explicit water at pH 2 to 12, with surface charge set from a deprotonation model, and with NaCl or $CaCl_2$ added at pH 10. Neutral particles showed no net drift in separation; partly charged particles (pH 7, 8) drifted apart by roughly 0.4 Å over about 50 ps; fully charged particles (pH 10, 12) drifted less (about 0.1 Å) while Na^+ accumulated more strongly at the surface. Salt runs showed no resolvable trend. These are single short runs and the results are qualitative.

1. Motivation

Dispersion versus aggregation of fine silica in water is set by surface charge (pH) and ionic screening, which governs slip rheology and packing in ceramic processing. It is directly relevant to **refractory castables**, where silica sol and microsilica are used as binder and fines and gel or deflocculate depending on pH and on Ca^{2+} released by calcium aluminate cement (pore-water pH 11–12). This study is a molecular-scale model of that mechanism at 300 K in water; it does not address high-temperature refractory properties.

2. Method

Spheres of 3 nm were cut from α-quartz and capped with silanol groups (1497 atoms, 234 surface Si–OH each, about 28 nm^2). The fraction of deprotonated sites was 1/(1 + 10^(pKa − pH)) with pKa 6.8, each carrying −1.525 e and balanced by one Na^+. Two particles were placed 40 Å apart in a periodic box (about 83 × 42 × 41 Å) filled with SPC water (14,600–16,000 atoms). The CLAYFF force field was used with SHAKE, 10 Å cutoff and PPPM. After minimisation and a 20 ps Langevin warm-up, 35–55 ps of NVT at 300 K (1 fs step) was run for each condition, tracking the distance between the two centres of mass.

pH	Deprotonated / NP	Charge / NP (e)	Net drift in separation
2, 4	0 of 234	0	≈ 0 (pH 4 identical to pH 2)
7	143 (61%)	−143	≈ +0.36 Å
8	220 (94%)	−220	≈ +0.4 Å
10, 12	234 (100%)	−234	≈ +0.08 Å (pH 12 reproduces pH 10)

Table 1. Charge states and observed drift. The model saturates, so the six pH values give four distinct charge states.

3. Results

Figure 1 shows that the first Na^+ peak at the surface is taller at pH 10 (0.038) than at pH 7 (0.020), consistent with stronger counterion binding, which can reduce the effective repulsion between particles. Figure 2 (next page) shows separation against time: the neutral particles at pH 2 oscillate about 40.4 Å without drift, at pH 7 the separation rises from about 40.7 to 41.1 Å, and at pH 10 it stays within 40.46–40.74 Å.

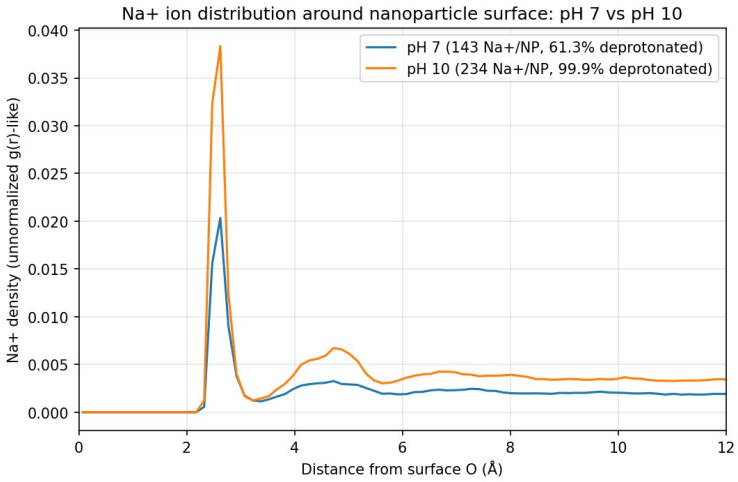


Figure 1. Na^+ density versus distance from surface oxygens, pH 7 and pH 10 (unnormalised).

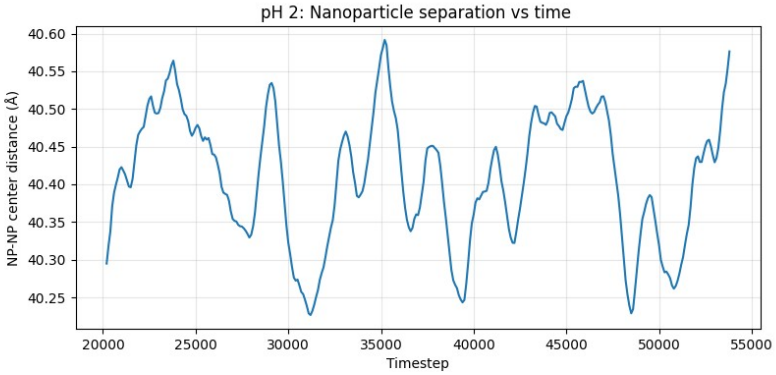


Figure 2a. pH 2: neutral surface, no net drift.

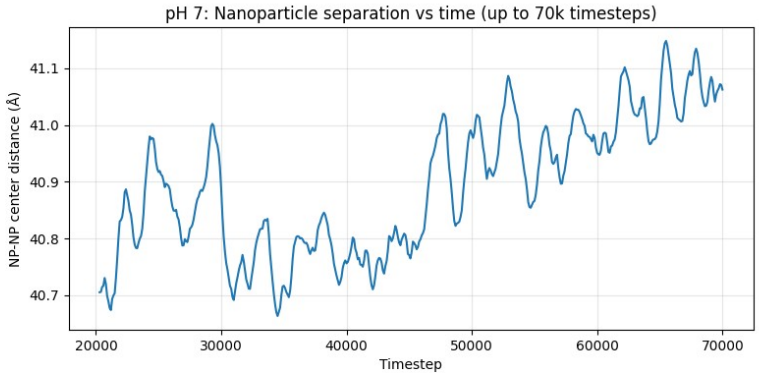


Figure 2b. pH 7: outward drift of about +0.36 Å.

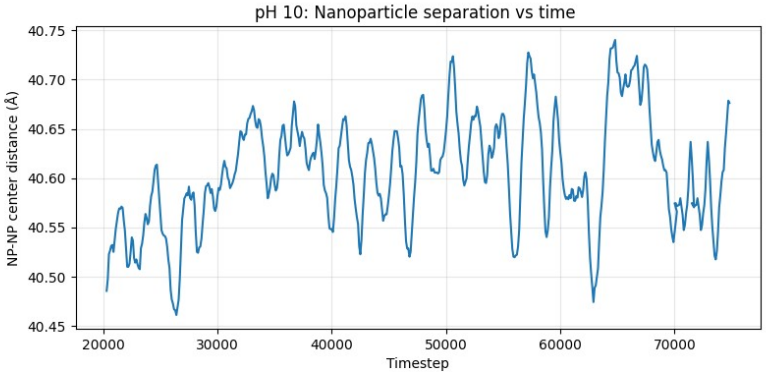


Figure 2c. pH 10: fully charged, small drift (about +0.08 Å).

Salt (pH 10). NaCl and $CaCl_2$ at 0.05 and 0.3 M (3 and 21 ion pairs; 25 ps each) all stay within 40.3–41.0 Å and overlap, with no consistent dependence on concentration or valence. The 468 neutralising Na^+ already give a background of several molar, so the added salt changes the screening very little in this model.

4. Limitations

- Surface charge is about 8 e/nm², several times higher than real silica, and the box is small (about 12 Å to the periodic image in y and z).
- Each condition is one 35–55 ps run; drifts of 0.1–0.4 Å are comparable to the 0.2–0.4 Å thermal fluctuation, so no error bars can be given.
- Charge comes from a fixed pKa; real silica stays stable near pH 2 through hydration forces that this model does not represent. The pH 8 value cannot be separated from pH 7 and pH 10.

5. Conclusion

Surface charge produces a small outward drift between nanoparticles, and Na^+ binding is a plausible reason why full charge does not increase it. Quantitative conclusions need a realistic charge density, a larger box, free-energy (PMF) calculations and replicas.